MAT3032 ANSWERS TO EXERCISES 1.1

1. (a) As H and J are subgroups, $e \in H$ and $e \in J$, so $e \in H \cap J$.
   Thus $H \cap J$ is non-empty. It is clearly a subset of G.
   Let $a, b \in H \cap J$. Then $a \in H$ and $b \in H$ so, as H is a group, $ab^{-1} \in H$.
   Also $a \in J$ and $b \in J$ so, as J is a group, $ab^{-1} \in J$.
   Hence $ab^{-1} \in H \cap J$ for all $a, b \in H \cap J$, so by the subgroup test $H \cap J$ is a subgroup of G.
   $H \cap J$ is thus a group, under the operation in G. It is contained in H and in J, so is a subgroup of both of them.

   (b) (i) If $|H| = 4, |J| = 6$ then $|H \cap J| = 1$ or 2.
   (ii) If $|H| = |J| = 7$ then $|H \cap J| = 1$ or 7 (in the latter case $H = J$).
   (iii) By (a) and Lagrange's theorem, $|H \cap J|$ divides both $|H|$ and $|J|$. If these are coprime then $|H \cap J| = 1$, i.e. $H \cap J$ is a trivial group.

2. (a) $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = e$, and similarly $(b^{-1}a^{-1})(ab) = e$, so $b^{-1}a^{-1}$ is the inverse of ab.

   (b) For all $a, b \in G$ we have $ab \in G$, so $(ab)^2 = e$. Thus $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$, so G is abelian.

3. $\phi : a \mapsto a^2$ is a homomorphism $\Leftrightarrow \phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$
   $\Leftrightarrow (ab)^2 = a^2b^2$ for all $a, b \in G \quad \Leftrightarrow \quad abab = a^2b^2$ for all $a, b \in G$
   $\Leftrightarrow a^{-1}(abab)b^{-1} = a^{-1}(a^2b^2)b^{-1}$ for all $a, b \in G$
   $\Leftrightarrow ba = ab$ for all $a, b \in G$, i.e. G is abelian.

4. (a) $(bab^{-1})^1 = bab^{-1} = ba^1b^{-1}$, so the result holds when $n = 1$.
   Assume it holds for $n = k$, so $(bab^{-1})^k = ba^kb^{-1}$.
   Then $(bab^{-1})^{k+1} = (bab^{-1})^k(bab^{-1}) = (ba^kb^{-1})(bab^{-1}) = ba^k(b^{-1}b)ab^{-1}$
   $= ba^keab^{-1} = ba^{k+1}b^{-1}$, so the result holds when $n = k + 1$.
   Hence by induction it is true for all $n \in \mathbb{N}$.

   (b) $b^1a^1b^{-1} = bab^{-1} = a^r = a^{(r^1)}$, so the result holds when $n = 1$.
   Assume it holds for $n = k$, so $b^kab^{-k} = a^{(r^k)}$.
   Then $b^{k+1}ab^{-(k+1)} = b(b^kab^{-k})b^{-1} = ba^{(r^k)}b^{-1} = (bab^{-1})^{(r^k)}$ (by part (a))
   $= (a^r)^{(r^k)} = a^{(rr^k)} = a^{(r^{k+1})}$, so the result holds when $n = k + 1$.
   Hence by induction it is true for all $n \in \mathbb{N}$.

5. For all $x, y \in G$ we have $\gamma_a(xy) = a(xy)a^{-1} = (axa^{-1})(aya^{-1}) = \gamma_a(x)\gamma_a(y)$, so γ_a is a group homomorphism. If $\gamma_a(x) = e$ (the identity of G) then $axa^{-1} = e$, so $x = a^{-1}ea = e$. Hence $\operatorname{Ker} \gamma_a = \{e\}$, so γ_a is injective.

   Given any $y \in G$, we have $y = a(a^{-1}ya)a^{-1} = \gamma_a(a^{-1}ya)$. As $a^{-1}ya \in G$, $y \in \gamma_a(G)$. Thus $\gamma_a(G) = G$, so γ_a is surjective. Hence γ_a is an automorphism.

   (Note: if G is assumed to be finite then an injective map from G to G must be bijective. However, this is not the case if G is infinite, e.g. $\phi : x \mapsto 2x$ is an injective but not surjective homomorphism from $(\mathbb{Z}, +)$ into $(\mathbb{Z}, +)$.)

6. Let $H = \{a \in G \;:\; \phi(a) = a\}$. By definition, H is a subset of G. As $\phi(e) = e$, we have $e \in H$ so H is non-empty.

   Let $a, b \in H$, so $\phi(a) = a, \phi(b) = b$. Then $\phi(ab^{-1}) = \phi(a)\phi(b^{-1}) = \phi(a)(\phi(b))^{-1} = ab^{-1}$, so $ab^{-1} \in H$. Thus H is a subgroup of G.

7.

	e	g	g^2	g^3	g^4
e	e	g	g^2	g^3	g^4
g	g	g^2	g^3	g^4	e
g^2	g^2	g^3	g^4	e	g
g^3	g^3	g^4	e	g	g^2
g^4	g^4	e	g	g^2	g^3

	e	g	g^2	g^3	g^4	g^5
e	e	g	g^2	g^3	g^4	g^5
g	g	g^2	g^3	g^4	g^5	e
g^2	g^2	g^3	g^4	g^5	e	g
g^3	g^3	g^4	g^5	e	g	g^2
g^4	g^4	g^5	e	g	g^2	g^3
g^5	g^5	e	g	g^2	g^3	g^4

C_5 has subgroups $\{e\}$, generated by e, and C_5, generated by any of $g, \ldots, g^4$.

C_6 has subgroups $\{e\}$, generated by e, $\{e, g^3\}$, generated by g^3; $\{e, g^2, g^4\}$, generated by g^2 (or g^4) and C_6, generated by g (or g^5).

C_7 has subgroups $\{e\}$, generated by e, and C_7, generated by any of $g, \ldots, g^6$.

C_8 has subgroups $\{e\}$, generated by e; $\{e, g^4\}$, generated by g^4; $\{e, g^2, g^4, g^6\}$, generated by g^2 (or g^6) and C_8, generated by g (or any of g^3, g^5, g^7).

C_9 has subgroups $\{e\}$, generated by e; $\{e, g^3, g^6\}$, generated by g^3 (or g^6) and C_9, generated by g (or any of g^2, g^4, g^5, g^7, g^8).

8. Each element which is *not* self-inverse can be paired with its inverse, leaving an even number of self-inverse elements. Of these, e has order 1 so there is an odd number of elements with order 2.

   Thus every group of order 4 has either only one element of order 2 in which case (as only e has order 1) the other two elements must have order 4, so the group is cyclic; or it has three elements of order 2, in which case it is isomorphic to V.

9. $84 = 2^2 \times 3 \times 7$ so $\varphi(84) = 84 \times \dfrac{1}{2} \times \dfrac{2}{3} \times \dfrac{6}{7} = 24$.

   Thus there are 24 distinct generators of C_{84} and 24 units in $(\mathbb{Z}_{84}, \times_{84})$, i.e. $|U_{84}| = 24$.

10. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 8 & 6 & 2 & 1 & 7 \end{pmatrix} = (1\ \ 4\ \ 8\ \ 7)(2\ \ 5\ \ 6) = (1\ \ 7)(1\ \ 8)(1\ \ 4)(2\ \ 6)(2\ \ 5)$.
    This is an odd permutation, so it is not an element of A_8.

11. Let $\sigma, \tau \in S_n$ be products of p and q transpositions respectively. Then $\sigma\tau$ is a product of $p + q$ transpositions. We have $\operatorname{sgn}(\sigma) = (-1)^p$, $\operatorname{sgn}(\tau) = (-1)^q$ and $\operatorname{sgn}(\sigma\tau) = (-1)^{p+q} = (-1)^p(-1)^q = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$. Thus sgn is a group homomorphism.

    For all $n \geq 2$ there are both even and odd permutations in S_n, so the image of sgn is $\{-1, 1\}$. The kernel consists of those elements of S_n which map to 1 (the identity in $\mathbb{R}^*$), so $\operatorname{Ker}(\operatorname{sgn}) = A_n$.

12. The subgroups $\left\{ \begin{pmatrix} u & a \\ 0 & 1 \end{pmatrix} : a \in \mathbb{Z}_n,\ u \in U_n \right\}$, $\left\{ \begin{pmatrix} 1 & a \\ 0 & v \end{pmatrix} : a \in \mathbb{Z}_n,\ v \in U_n \right\}$ and $\left\{ \begin{pmatrix} u & a \\ 0 & u \end{pmatrix} : a \in \mathbb{Z}_n,\ u \in U_n \right\}$ each have order $n\,\varphi(n)$. (n choices for a, $\varphi(n)$ choices for u or v.) The last of these is abelian.

    For the first subgroup $\begin{pmatrix} u & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} u' & a' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} uu' & ua' + a \\ 0 & 1 \end{pmatrix}$, while

    $\begin{pmatrix} u' & a' \\ 0 & 1 \end{pmatrix}\begin{pmatrix} u & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} u'u & u'a + a' \\ 0 & 1 \end{pmatrix}$. When $n = 2$ the only unit is 1 so $u' = u = 1$ and the above two matrices are equal. Otherwise $ua' + a \neq u'a + a'$ in general, so this subgroup is non-abelian. Likewise for the second one.

    The subgroup $\left\{ \begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix} : u, v \in U_n \right\}$ has order $\varphi(n)^2$ and is abelian.

The subgroups $\left\{\begin{pmatrix} 1 & 0 \\ 0 & v \end{pmatrix} \ : \ a \in \mathbb{Z}_n,\ v \in U_n\right\}$, $\left\{\begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} \ : \ a \in \mathbb{Z}_n,\ u \in U_n\right\}$ and $\{u\mathrm{I}_2 \ : \ u \in U_n\}$ each have order $\varphi(n)$ and are abelian.

(a) When n is prime, $|R_n| = n(n-1)^2$ and the subgroups above have orders $n(n-1)$, $(n-1)^2$, $n-1$.

(b) When $n = 4$, $|R_n| = 16$ and the subgroups above have orders 8, 4, 2.

13. (a) Suppose G and H are abelian. Let $(g_1, h_1), (g_2, h_2) \in G\times H$. Then $(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2) = (g_2g_1, h_2h_1) = (g_2, h_2)(g_1, h_1)$, so $G \times H$ is abelian.

(b) $\phi((g_1, h_1)(g_2, h_2)) = \phi(g_1g_2, h_1h_2) = (h_1h_2, g_1g_2) = (h_1, g_1)(h_2, g_2)$ $= \phi(g_1, h_1)\phi(g_2, h_2)$, so ϕ is a group homomorphism.

$\operatorname{Ker}\phi = \{(g, h) \in G \times H \ : \ (h, g) = (e_2, e_1)\} = \{(e_1, e_2)\}$. Thus $\operatorname{Ker}\phi$ contains only the identity of $G \times H$, so ϕ is injective.

$\phi(G \times H) = \{(h, g) \ : \ h \in H, g \in G\} = H \times G$, so ϕ is surjective.

Hence ϕ is a group isomorphism, so $G \times H \cong H \times G$.

(c) $(e_1, e_2) \in G \times \{e_2\}$ so $G \times \{e_2\}$ is non-empty. It is clearly contained in $G \times H$. Let $a = (g_1, e_2), b = (g_2, e_2) \in G \times \{e_2\}$. Then $ab^{-1} = (g_1, e_2)({g_2}^{-1}, e_2) = (g_1{g_2}^{-1}, e_2) \in G \times \{e_2\}$, so $G \times \{e_2\}$ is a subgroup of $G \times H$.

Define $\psi : G\times\{e_2\} \to G$ by $\psi : (g, e_2) \mapsto g$. Then $\psi((g, e_2)(h, e_2)) = \psi(gh, e_2) = gh = \psi(g, e_2)\psi(h, e_2)$, so ψ is a group homomorphism.

If $\psi(g, e_2) = e_1$ then $(g, e_2) = (e_1, e_2)$, the identity in $G \times \{e_2\}$, so ψ is injective. $\psi(G\times\{e_2\}) = G$ so ψ is surjective. Hence ψ is an isomorphism, so $G\times\{e_2\} \cong G$.

(d) $C_2 \times \{e_2\} \cong C_2$, $\quad \{e_1\} \times C_5 \cong C_5$.

14. $(e, e) \in \Delta$, so Δ is non-empty. It is clearly a subset of $G \times G$.

Let $x = (a, a), y = (b, b) \in \Delta$.

Then $xy^{-1} = (a, a)(b^{-1}, b^{-1}) = (ab^{-1}, ab^{-1}) \in \Delta$.

Thus Δ is a subgroup of $G \times G$.

Let H_1, H_2 be subgroups of G. If both are trivial then $H_1 \times H_2 = \{(e, e)\} \subsetneq \Delta$.

Otherwise there is an element $a \neq e$ in one subgroup so (e, a) or (a, e) is in $H_1 \times H_2$ but not in Δ. Thus Δ does not have the form $H_1 \times H_2$.